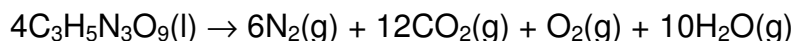


- 1 (a) Alfred Noble discovered that an explosive could be made from nitroglycerin (a liquid which on its own readily detonates) mixed with a substance called Kieselguhr (diatomaceous earth). The equation for the explosion is:



This discovery contributed to his enormous wealth, part of which he used to establish the Nobel Prizes.

- (i) Given that the volume of gases produced at 500°C and 1 atm was 83.0 dm^3 . Calculate the mass of nitroglycerin that had decomposed.

[2]

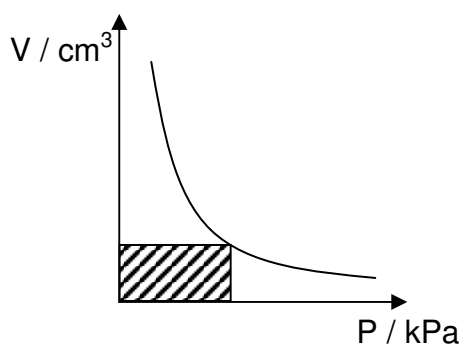
- (ii) Apart from the great amount of heat energy released, why would this reaction cause extensive damage if it is carried out in a closed area?

[1]

- (b) Among the gaseous products formed at 500°C , Alfred Noble studied the behaviour of these gases.
State and explain which of these gaseous products behave most ideally.

[2]

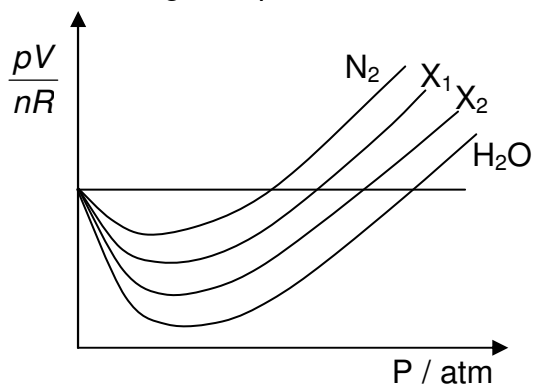
- (c) The graph below was mapped out by Alfred Noble. He related the behaviour of the most ideal gas at constant temperature.



Given that the area of the shaded region = $5.82 \times 10^4 \text{ cm}^3 \text{ kPa}$, calculate the pressure exerted by the gas occupying a volume of 112 cm^3 .

[2]

- (d) Alfred Noble continued to study the behaviour of the gases. He showed a graph of compressibility curves of the gases produced from nitroglycerine.



Identify the gases labelled X_1 and X_2 .

X_1 -

X_2 -

[1]

[Total: 8]

- 2 (a) From the list of elements given below, select the one which has the stated property. Each element may be used once, more than once or not at all.

Elements: aluminium, carbon, silicon, magnesium, sodium, chlorine

Property		Element
(i)	Forms a chloride which produces an acidic solution in water	
(ii)	Forms an amphoteric oxide	
(iii)	Has an oxidation state of +7 in one of its oxides	
(iv)	Forms an ionic oxide of empirical formula XO	
(v)	Forms a giant covalent oxide	
(vi)	Forms a giant molecular structure that can conduct electricity	

[3]

- (b) An effective way of getting rid of any unpleasant taste and odour in water is the use of a synthetic yellowish green gas, chlorine dioxide, ClO_2 . This gas is synthesised using sodium chlorate(V) and hydrogen peroxide in an acidic solution.

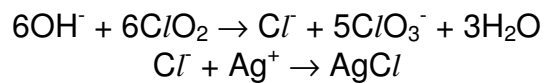
- (i) Give the dot-cross diagram of ClO_2 and predict its shape.

[2]

- (ii) Write a balanced ionic equation for the reaction between chlorate(V) ions and hydrogen peroxide.

[1]

- (c) It has been found that when water becomes alkaline, chlorine dioxide undergoes disproportionation forming Cl^- and ClO_3^- . A precipitate, AgCl is formed when 20 cm^3 of 0.05 mol dm^{-3} of $\text{AgNO}_3(\text{aq})$ is added.



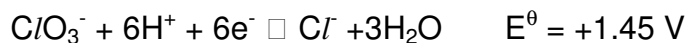
- (i) Explain what disproportionation means.

[1]

- (ii) Using the above information, calculate the volume of ClO_2 disproportionated at room temperature and pressure.

[2]

- (d) The standard electrode potential of chlorate(V) ions, ClO_3^- is given below.



- (i) Draw a fully labelled diagram to show how you would determine the standard electrode potential of $\text{ClO}_3^-/\text{Cl}^-$ electrode system.

[3]

- (ii) By using the E^θ value in (d), predict the reaction, if any, of ClO_3^- ions in acid solution with aqueous iron(II) ions.

[2]

- (iii) A student accidentally added an excess of sodium hydroxide to the above half-cells in d(ii). Suggest what will happen to the electrode potential of this cell.

[2]

[Total: 16]

- 3** Citric acid, RCO_2H and its sodium salt, RCO_2Na are often used as a buffer in foodstuffs.

Given: K_a of $\text{RCO}_2\text{H} = 7.4 \times 10^{-4} \text{ mol dm}^{-3}$ and K_b of $\text{RCO}_2\text{Na} = 1.4 \times 10^{-11} \text{ mol dm}^{-3}$

- (a) (i)** Calculate the pH at which the citric acid/citrate buffer functions at its maximum buffering capacity.

[1]

- (ii)** Write equations to show how the citric acid/citrate buffer system regulate the acidity on the addition of H^+ and OH^- ions.

[2]

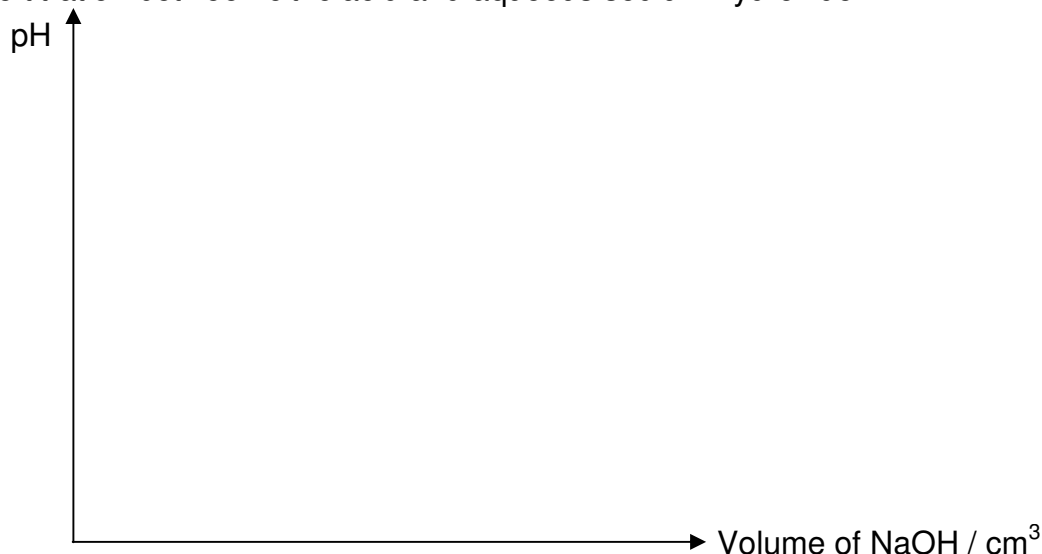
- (b)** 30.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ of citric acid is added to 20.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ of sodium hydroxide. Calculate the pH of the resulting mixture.

[2]

- (c)** Calculate the pH of 0.05 mol dm^{-3} of aqueous sodium citrate solution.

[2]

- (d) Using the pH values calculated from (a) to (c), sketch the shape of the pH curve of the titration between citric acid and aqueous sodium hydroxide.



[3]

[Total: 10]

- 4 Vitamin B₁₂ is found to contain cobalt. It is required for the formation of the nucleic acids DNA and RNA which embody the genetic code. The following passage is concerned with the preparation of a complex of cobalt.

Dissolve 12 g of potassium ethandioate, K₂C₂O₄ and 4 g of ethanedioic acid, H₂C₂O₄ in 80 cm³ of boiling water. Add 4 g of cobalt(III) carbonate slowly. A deep purple solution will be obtained and it contained the complex ion [Co(C₂O₄)₂]²⁻. This solution should be cooled to 40 °C. Add slowly to this solution 10 g of lead(IV) oxide, stirring continuously. Carefully add, drop by drop, 4 cm³ of moderately concentrated ethanoic acid. The solution should now turn dark green. Allow crystallisation to take place in the dark so as to obtain the crystals of the complex K₃[Co(C₂O₄)₃].

- (a) Give the oxidation state of cobalt in the purple complex ion and the coordination number of the final complex ion.

Oxidation number of cobalt in the purple complex -

Coordination number of the final complex ion -

[2]

- (b) What is the function of lead(IV) oxide?

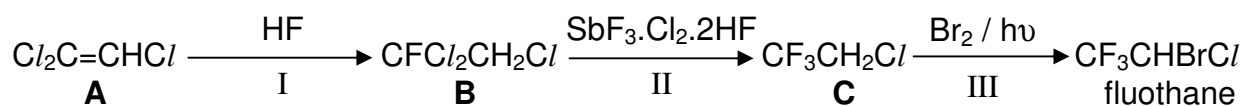
[1]

(c) Explain why both of the complexes of cobalt are coloured.

[3]

[Total: 6]

5 The anaesthetic, Fluothane, is made by the following route.



(a) Give the name of the type of mechanism involved in

Step I -

Step III -

[2]

(b) (i) $h\nu$ signifies a certain reaction condition for step III. Write an equation to illustrate the role of $h\nu$ in step III.

..... [1]

(ii) What gas is produced as well as Fluothane in Step III?

..... [1]

(iii) Give the hybridised state of carbon atoms in Fluothane and suggest a value for the F-C-F bond angle.

..... [2]

- (c) 9.0 g of Fluothane was obtained from 10.0 g of compound **C**. What was the percentage yield for step III?
(M_r for compound C = 118.5; A_r , Br = 80.0; A_r , H = 1.0)

[2]

- (d) Using data from the Data Booklet, calculate the standard bond enthalpy of the C-F bond, given that $\Delta H = -40 \text{ kJ mol}^{-1}$ for step I.

[2]

- (e) Compound **C** can be converted into $\text{CF}_3\text{CH}_2\text{OH}$.
(i) Suggest a reagent for this conversion.

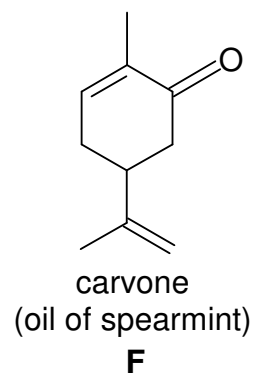
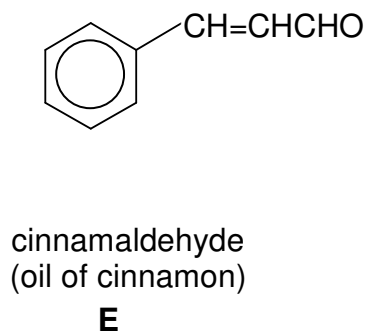
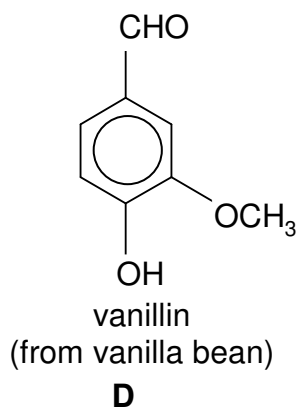
[1]

- (ii) Suggest a mechanism for this conversion, by means of equations. In your answer, indicate any relevant charges and movement of electrons.

[2]

[Total: 13]

- 6 Three important naturally-occurring compounds used as flavourings in the food industry are shown below:



- (a) State which compound can be identified by its reaction with neutral iron(III) chloride solution. State the result you would expect to see.

[1]

- (b) Suggest the structural formula of the organic product formed, if any, when HBr is added to the above compounds.

[2]

- (c) Only one of the compounds can react with sodium. Give an equation for this reaction.

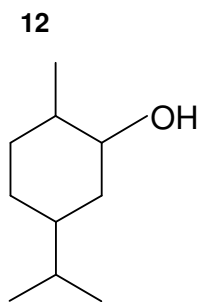
[1]

- (d) Compound **F** is heated with hot concentrated manganate(VII) ions. Draw the displayed formula of the organic compound formed.

[2]

(e)

Compound **F** can be converted to



Give the reagent(s) and conditions for this conversion.

[1]

[Total: 7]

End of Paper